

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 6

For the tutorial in Week 7 on 29 February, let us discuss

- Ex. 8.2.11, 8.2.22, 8.2.24, 8.2.25, 8.2.26, additional exercises.

Ex. 8.2.11. Find the solution to $a_n = 2a_{n-1} + 5a_{n-2} - 6a_{n-3}$ with $a_0 = 7, a_1 = -4, a_2 = 8$.

Ex. 8.2.22.

- (a) Find all solutions of the recurrence relation $a_n = 2a_{n-1} + 2n^2$ for $n \geq 2$.
- (b) Find the solution of the recurrence relation in part (a) with initial condition $a_1 = 4$.

Ex. 8.2.24.

- (a) Find all solutions of the recurrence relation $a_n = -5a_{n-1} - 6a_{n-2} + 42 \cdot 4^n$.
- (b) Find the solution of the recurrence relation in part (a) with initial conditions $a_1 = 56$ and $a_2 = 278$.

Ex. 8.2.25. Find the solution of the recurrence relation $a_n = 4a_{n-1} - 3a_{n-2} + 2^n + n + 3$ with $a_0 = 1$ and $a_1 = 4$.

(*Hint:* Find $a_n^{(p1)}$ which takes care of the exponential part, i.e., a particular solution of $4a_{n-1} - 3a_{n-2} + 2^n$. Next, find $a_n^{(p2)}$ which takes care of the polynomial part, i.e., a particular solution of $4a_{n-1} - 3a_{n-2} + n + 3$. Finally, add them up, $a_n^{(p)} = a_n^{(p1)} + a_n^{(p2)}$.)

Ex. 8.2.26. Let a_n be the sum of the first n perfect squares, that is $a_n = \sum_{k=1}^n k^2$. Using linear non-homogeneous recurrence relations to determine a formula for a_n .

(*Hint:* observe the relation between a_n and a_{n-1} , i.e. $a_n = a_{n-1} + ?$)

Additional Exercise 1. Solve the recurrence relation $a_n = a_{n-1}^2/a_{n-2}$ if $a_0 = 1$ and $a_1 = 2$. (*Hint:* take logarithms of both sides to obtain a recurrence relation for the sequence $\{\log a_n\}$. Alternatively, one can make an observation on the ratio a_n/a_{n-1} by working out the values $a_1/a_0, a_2/a_1, a_3/a_2$, etc.)

Additional Exercise 2. Show that the Fibonacci numbers satisfy the recurrence relation $f_n = 5f_{n-4} + 3f_{n-5}$ for $n = 5, 6, 7, \dots$ together with the initial conditions $f_0 = 0, f_1 = 1, f_2 = 1, f_3 = 2, f_4 = 3$.

Additional exercise 3. Suppose that the sequences $(s_0, s_1, s_2, \dots)$ and $(t_0, t_1, t_2, \dots)$ both satisfy the same linear homogeneous recurrence relation with constant coefficients: for all integers $k \geq 2$, $s_k = 5s_{k-1} - 4s_{k-2}$ and $t_k = 5t_{k-1} - 4t_{k-2}$. Define a new sequence $(u_0, u_1, u_2, \dots)$ where $u_i = 2s_i + 3t_i$. Show that $(u_0, u_1, u_2, \dots)$ also satisfies the same relation (i.e. $u_k = 5u_{k-1} - 4u_{k-2}$).

The following are optional exercises. We will not be discussing them during tutorial.

Ex. 8.2.14. Solve the recurrence relation $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$ with $a_0 = -5, a_1 = 4, a_2 = 88$.

Ex. 8.2.15. Solve the recurrence relation $a_n = -3a_{n-1} - 3a_{n-2} - a_{n-3}$ with $a_0 = 5, a_1 = -9, a_2 = 15$.

Ex. 8.2.17 and 8.2.18. What is the general form of the solutions of a linear homogeneous recurrence relation if its characteristic equation has roots

- (a) $1, 1, 1, 1, -2, -2, -2, 3, 3, -4$?
- (b) $-1, -1, -1, 2, 2, 5, 5, 7$?

Ex. 8.2.23.

- (a) Find all solutions of the recurrence relation $a_n = 2a_{n-1} + 3^n$ for $n \geq 2$.
- (b) Find the solution of the recurrence relation in part (a) with initial condition $a_1 = 5$.

Answer.

Ex. 8.2.11. $a_n = 5 + 3 \cdot (-2)^n - 3^n$ for $n \geq 0$.

Ex. 8.2.22. $a_n = h \cdot 2^n - 2n^2 - 8n - 12$ for $n \geq 2$; $a_n = 13 \cdot 2^n - 2n^2 - 8n - 12$ for $n \geq 1$.

Ex. 8.2.24. $a_n = h_1 \cdot (-2)^n + h_2 \cdot (-3)^n + 4^{n+2}$; $a_n = (-2)^n + 2 \cdot (-3)^n + 4^{n+2}$ for $n \geq 1$.

Ex. 8.2.25. $a_n = 1/8 + 39/8 \cdot 3^n - 2^{n+2} - n^2/4 - 5n/2$ for $n \geq 0$.

Ex. 8.2.26. $a_n = n(n+1)(2n+1)/6$ for $n \geq 1$.

Additional Exercise 1. $a_n = 2^n$ for $n \geq 0$.